

security number. Then ask him to estimate the number of dentists in Manhattan. You will find that by making him aware of the four-digit number, you elicit an estimate that is correlated with it.

We use reference points in our heads, say sales projections, and start building beliefs around them because less mental effort is needed to compare an idea to a reference point than to evaluate it in the absolute (*System 1* at work!). We cannot work without a point of reference.

So the introduction of a reference point in the forecaster's mind will work wonders. This is no different from a starting point in a bargaining episode: you open with high number ("I want a million for this house"); the bidder will answer "only eight-fifty"—the discussion will be determined by that initial level.

The Character of Prediction Errors

Like many biological variables, life expectancy is from Mediocristan, that is, it is subjected to mild randomness. It is not scalable, since the older we get, the less likely we are to live. In a developed country a newborn female is expected to die at around 79, according to insurance tables. When she reaches her 79th birthday, her life expectancy, assuming that she is in typical health, is another 10 years. At the age of 90, she should have another 4.7 years to go. At the age of 100, 2.5 years. At the age of 119, if she miraculously lives that long, she should have about nine months left. As she lives beyond the expected date of death, the number of additional years to go decreases. This illustrates the major property of random variables related to the bell curve. The conditional expectation of additional life drops as a person gets older.

With human projects and ventures we have another story. These are often scalable, as I said in [Chapter 3](#). With scalable variables, the ones from Extremistan, you will witness the exact opposite effect. Let's say a project is expected to terminate in 79 days, the same expectation in days as the newborn female has in years. On the 79th day, if the project is not finished, it will be expected to take another 25 days to complete. But on the 90th day, if the project is still not completed, it should have about 58 days to go. On the 100th, it should have 89 days to go. On the 119th, it should have an extra 149 days. On day 600, if the project is not done, you will be expected to need an extra 1,590 days. As you see, *the longer you wait, the longer you will be expected to wait*.

Let's say you are a refugee waiting for the return to your homeland. Each day that passes you are getting farther from, not closer to, the day of triumphal return. The same applies to the completion date of your next opera